INTEL.ai

Autonomous Company Research Agent *— AI Bootcamp Project*

LANGGRAPH | *RAG* | *N8N* | *REAL-TIME APIS*

Team: Anson, Szilard, Kinda

The Problem

Executives need fast, reliable company intelligence — but gathering it manually is slow and fragmented.

Financial Performance

Market data scattered across platforms

News & Developments

No single consolidated news feed

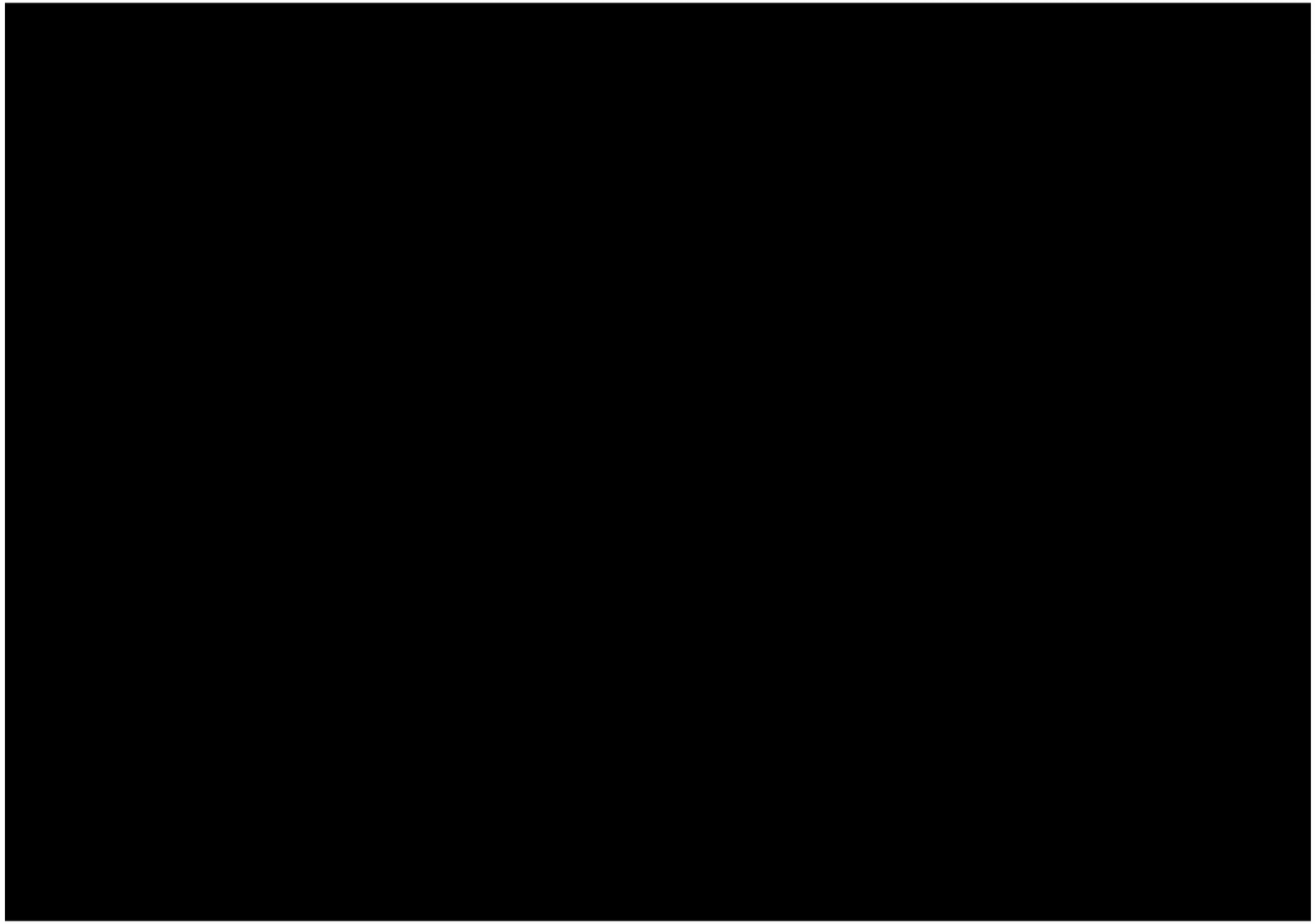
Competitive Positioning

Manual benchmarking is time-intensive

Strategic Risks

Signals buried in unstructured sources

Initial Approach – n8n Workflow



*The first system was built entirely in **n8n** — handling API orchestration, web search, news retrieval, financial data, and bundling results into a research package.*

*Complex workflow debugging revealed a need to separate **orchestration** from **AI reasoning** — leading to the current split architecture.*

→ API Orchestration

→ Data Retrieval & Bundling

→ AI Reasoning Layer (LangGraph)

Data Sources

Three real-time APIs feed the research pipeline — delivering structured search, news, and financial data at near-zero cost.

Serper API

Structured Google search results — company overviews, competitive context, and web intelligence.

Guardian API

Recent news articles — market developments, executive moves, and breaking industry coverage.

Alpha Vantage API

Live financial market data — stock price, volume, earnings, and key financial metrics.

$0.003

Avg. Cost Per Report

~$0.002–$0.005 per full intelligence report generated

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Live API Integrations

Serper, Guardian, and Alpha Vantage running in parallel

Real-time

Data Freshness

All data fetched live at query time — no stale cached results

Prototype System

The first working pipeline validated the core concept — company name in, structured intelligence report out.

The early prototype confirmed that the system could automatically synthesize multi-source research and generate coherent, structured reports — with no manual intervention at any stage.

Final AI Agent (LangGraph Reasoning Engine)

*The **LangGraph state machine** implements a **ReAct reasoning pattern** — each node performs a discrete reasoning step before passing state forward.*

Research Node

Queries and structures raw API data

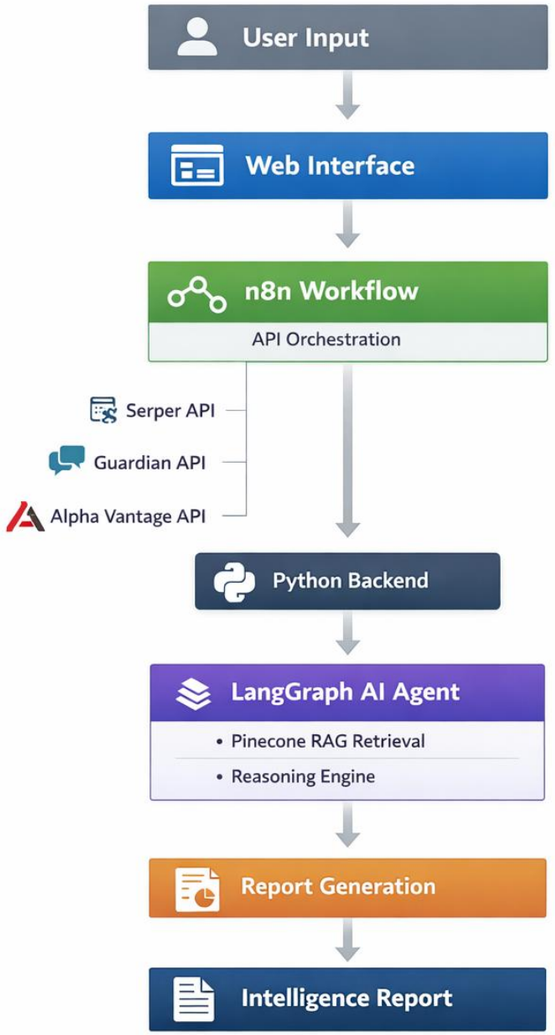
RAG Retrieval

Pinecone vector search for contextual grounding

Validation Loop

Auto-retries reasoning if output is incomplete

System Architecture



A clean separation of concerns across four layers:

User Interface

Web interface accepts company name and ticker symbol

n8n Orchestration

Coordinates API calls via Serper, Guardian, Alpha Vantage

Python Backend

Routes data between orchestration and AI agent layers

LangGraph AI Agent

ReAct reasoning + Pinecone RAG retrieval engine

Report Output

Structured intelligence report delivered to user

Live Demo

*Enter a **company name** and **ticker symbol**. The system handles everything else — automatically.*

Data Collection

Real-time API queries fire in parallel

AI Reasoning

LangGraph state machine processes and synthesizes

Intelligence Report

Executive Summary · Financials · News · Market Position · Strategic Assessment

Achievements

*A fully autonomous research agent — built from scratch in **4 days**.*

LangGraph + n8n Pipeline

ReAct reasoning state machine with full n8n orchestration layer

Pinecone RAG System

Context-aware retrieval grounding every report in verified knowledge

3 Live API Integrations

Serper, Guardian, and Alpha Vantage delivering real-time intelligence

Web Interface + Cloud

Deployed and accessible — full reports generated in seconds

Bottom line: *Full company intelligence reports — automatically generated, end-to-end — in seconds.*